

Generative AI meets Agent-Based Modelling

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Content

- Agent-Based Modelling (ABM)
 - What is ABM?
 - Examples of ABMs
- Use of LLMs throughout the ABM Study Lifecycle
 - Example 1: Using LLMs for ABM Design
 - Example 2: Generative Agent-Based Modelling

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Peer's ABM Blog



Exploring the dynamics of social systems

Image by Gemini Flash 2.5 (06/2025)

This is my blog on Agent-Based Modelling (ABM). It is simply a webpage I update from time to time. There is no subscription option. This blog is intended as a space for sharing news, discussions, and information related to my research on Computational ABM.

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Please note: These posts do not have any scientific rigour. They are simply exploratory pieces.

- [22/12/2025: 16 Special Issue on 'Generative AI meets Agent-Based Modelling and Simulation'](#)
- [22/12/2025: 15 LLM4ABM Special Interest Group \(SIG\) Established as an ESSA SIG](#)
- [14/12/2025: 14 LLM Support for Creating RAT-RS Reports \(2/2\): Quality & Reliability](#)
- [06/12/2025: 13 LLM Support for Creating RAT-RS Reports \(1/2\): Feasibility](#)
- [01/11/2025: 12 The Generative ABM Experiment \(1/3\): From Concept to First Prototype](#)
- [12/10/2025: 11 From Rules to Reasoning: Engineering LLM-Powered Agent-Based Models](#)



Agent-Based Modelling and Simulation

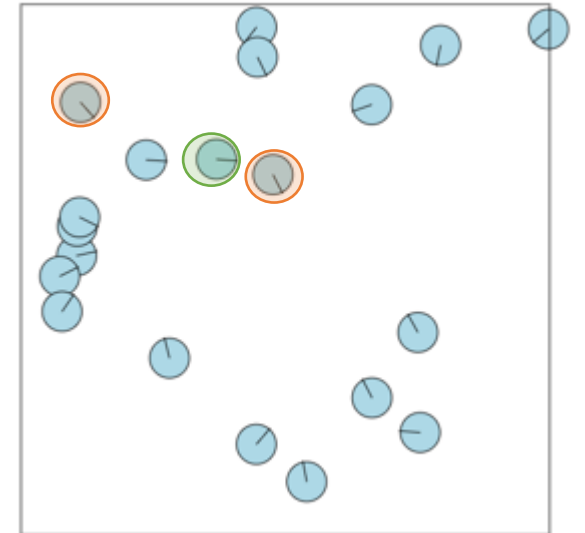
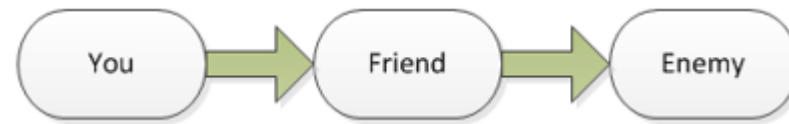
Heroes and Cowards Model [Wilensky and Rand 2015]

- Conceptual Definition

- The world consists of heroes and cowards
- A **hero** would always locate itself between a friend and enemy to shield one from the other

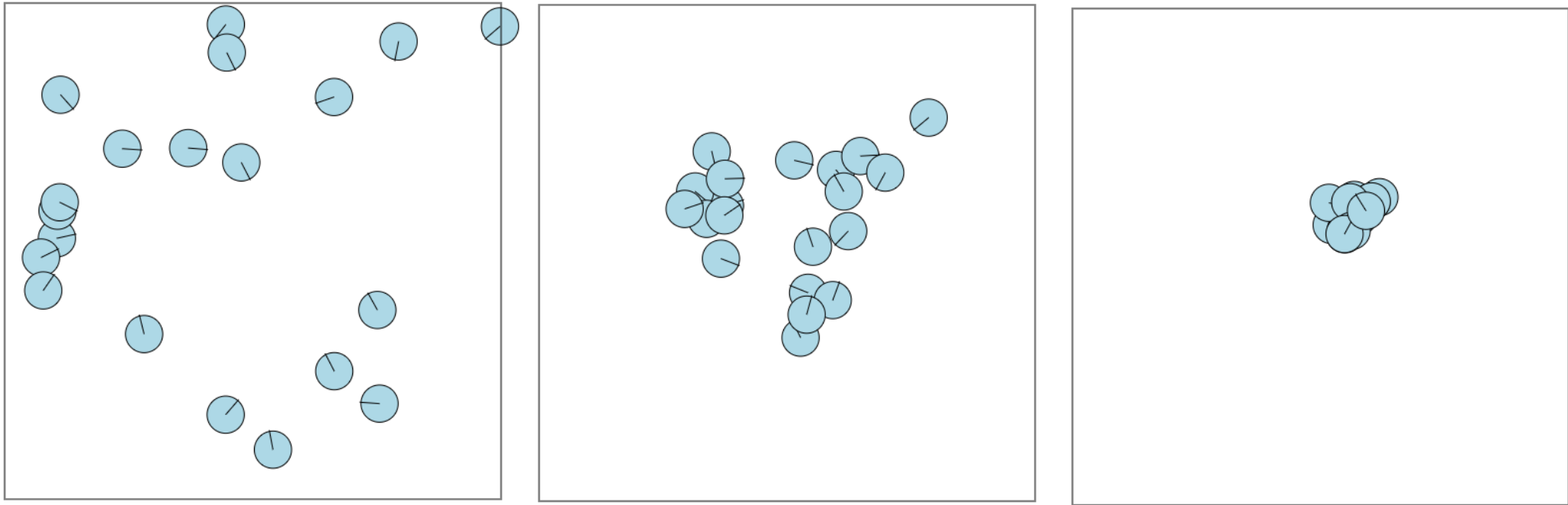


- A **coward** would always use a friend to shield itself from the enemy



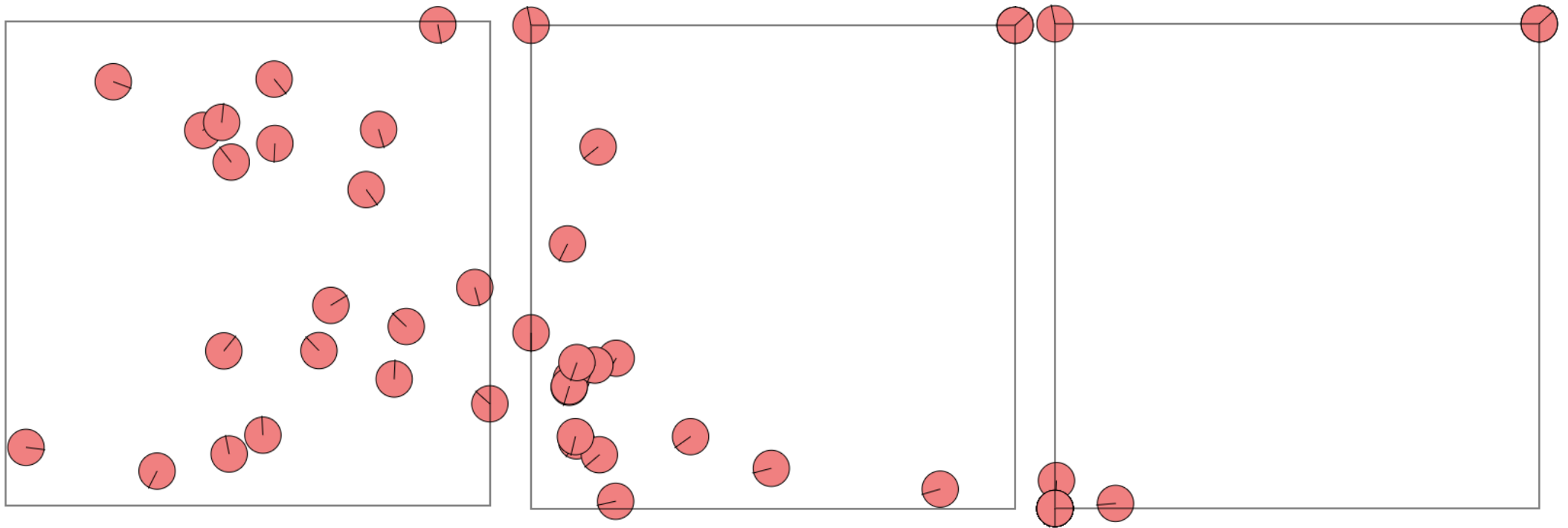
Heroes and Cowards

- Examining theoretical propositions through empirical testing: All heroes



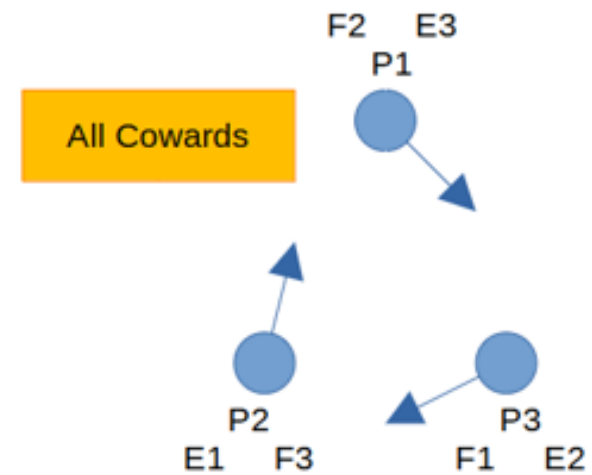
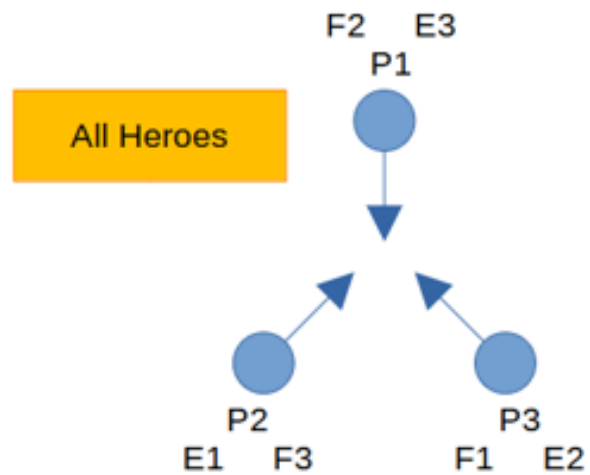
Heroes and Cowards

- Examining theoretical propositions through empirical testing: All cowards



Heroes and Cowards

P=Player
F=Friend
E=Enemy



Heroes and Cowards

- Sensitivity Analysis

- Has the proportion an impact?
- Has stochasticity an impact?
- Has the number of participants an impact?
- Are patterns dynamic?
- Does time play a role?

Once we have an implementation of the model, we can investigate all these questions very easily!



65% heroes 35% cowards - Software: AnyLogicPLE (<https://www.anylogic.com/>)

Agent-Based Modelling

- Agent-Based Modelling (formal definition)
 - A complex system is represented by a **collection of agents** that are programmed to **follow some behaviour rules**, and the **system properties emerge** from its constituent agent interactions



- Widely used in social science, epidemiology, economics, and ecology

Agent-Based Social Simulation (ABSS)

- ABSS applies ABM specifically to social systems – modelling people, groups, institutions and their interactions.
- Examples:
 - Social Cohesion: How groups form and maintain bonds
 - Social Norms: Rules that emerge without central authority
 - Cultural Dynamics: Spread of beliefs and behaviours
 - Spatial Patterns: How people organise in physical space

Building Agent-Based Models

- Schematic representation of an agent-based model

- Develop agent-based models
 - Define the problem
 - Identify active entities (agents)
 - Define their states and behaviour
 - Put them in an environment
 - Establish connections
 - Validate (bugs and plausibility)

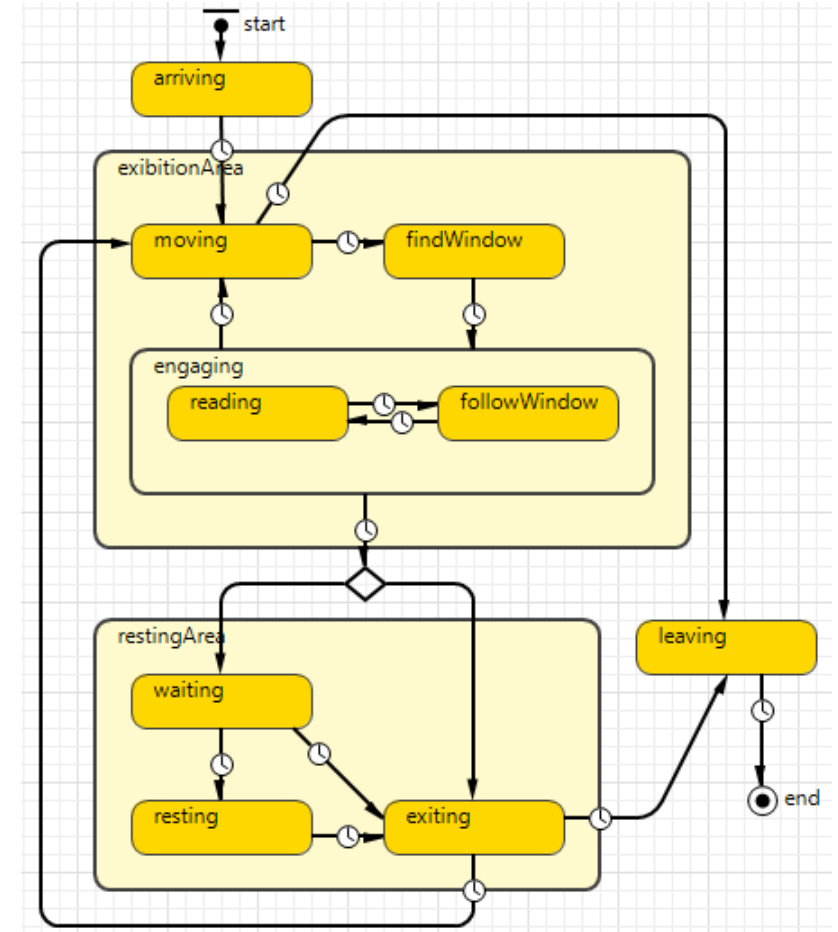
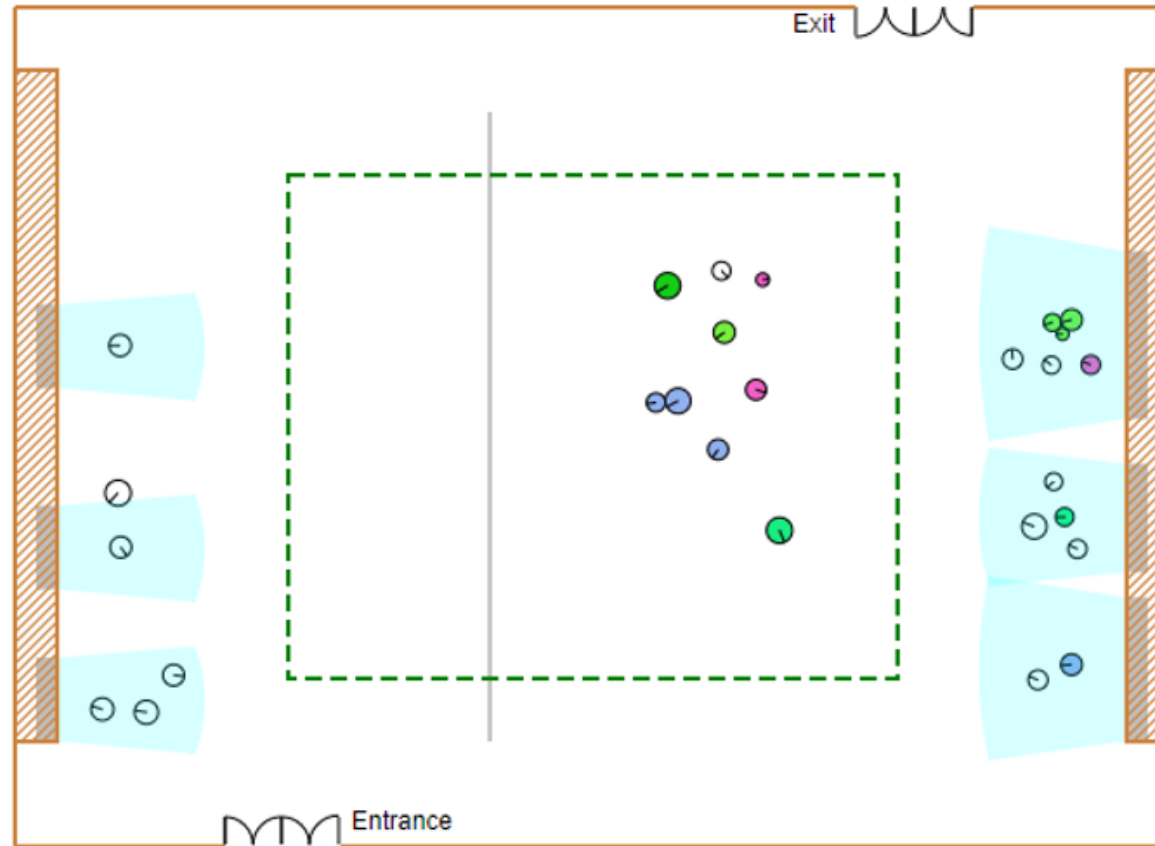


Ramadijah et al (2021)

Adaptive Architecture Simulation

Museum-Adaptiv-Wall*

- > AdaptiveWall
- > Group
- > Main
- > Person
- > Room
- > Screen
- > Simulation: Main
- > Run Configuration: Main
- > Database
- > Resources

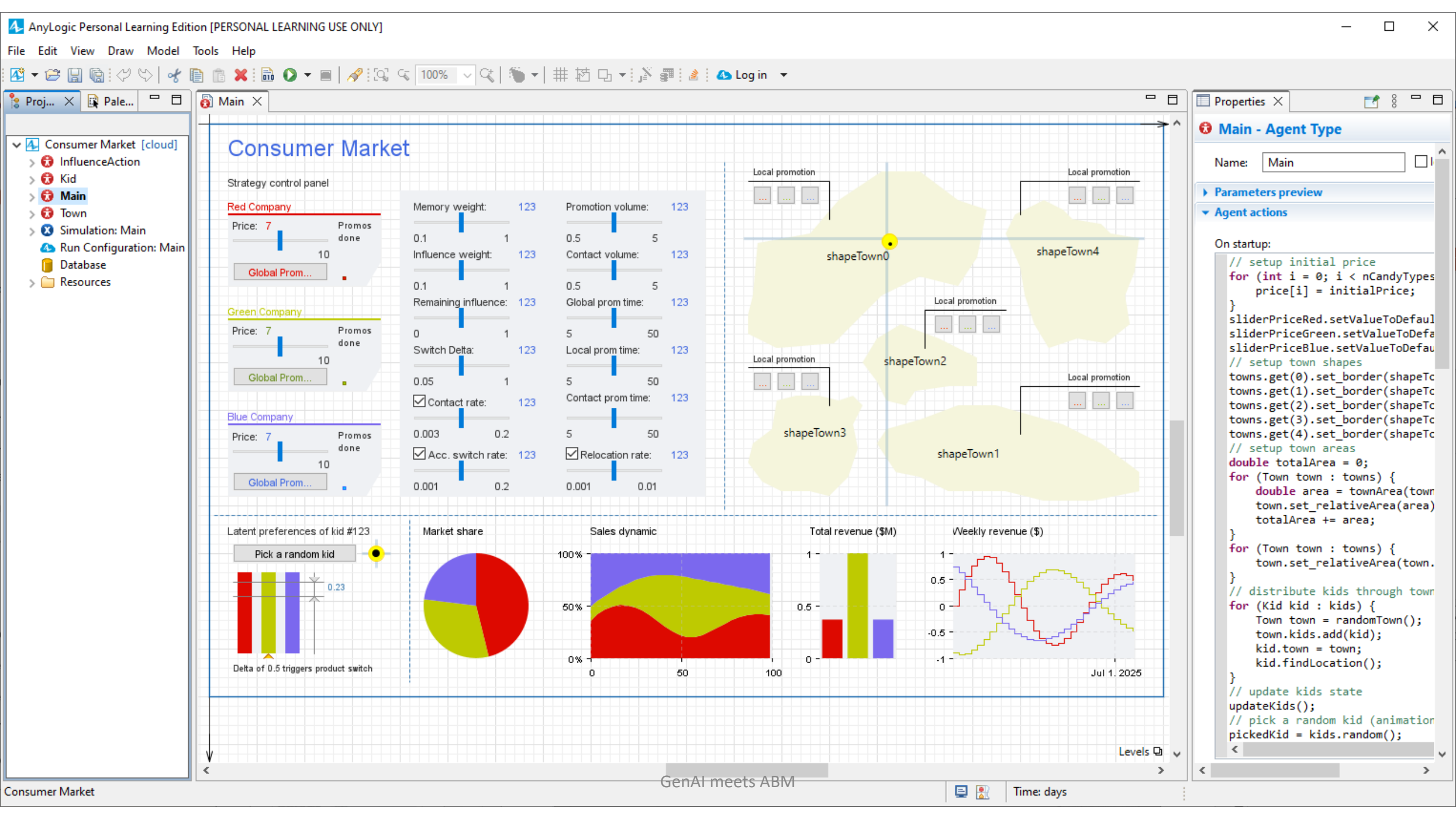




ABM Example

Dynamic Candy Market

Modelling Consumer Behaviour and Competition Among Kids




Consumer Market : Simulation - AnyLogic Personal Learning Edition

Consumer Market

marketplace / competition / consumer choice / brand loyalty / product switching

This is a simple marketplace and competition model. Kids live in five different towns and consume candies. There are three companies each producing one type of candies - red, green, or blue. Each kid has one favorite candy type but can switch to another type being influenced by other kids, promotion actions, or accidentally. Kids can move from one town to another and contact other kids in the same town.

This model is designed as a game. You can change the price for each candy type, do local (town-wide) or global promotion - and thus force the kids to change their preferences and purchase a particular type of candies. Companies market share and revenue charts are displayed.

?

Memory weight:

1

0.1

1

The weight of historical preferences in the current preference evaluation

Promotion volume:

1

0.5

5

The volume of promotion caused by a promotion action and affects the influence of a kid

Influence weight:

0.5

0.1

1

The weight of promotional and word-of-mouth influence in the current preference evaluation

Contact volume:

2

0.5

5

The volume of promotion caused by a contact between kids and affects the influence of a kid

Remaining influence:

0.2

0

1

The part of the influence that retains in the memory when the main effect ends

Global promotion time:

30

5

50

The duration in days shows how long the global promotion takes place

Switch Delta:

0.5

0.05

1

The critical preference difference between the currently preferred candy and another one that triggers the candy type switch

Local promotion time:

45

5

50

The duration in days shows how long the local promotion takes place

☒ Contact rate:

0.02

0.003

0.2

Number of contacts per day for a kid. Each contact results in an influence action (promotion of the favorite candy type)

Contact promotion time:

15

5

50

The duration in days shows how long the promotion caused by a contact between kids takes place

☒ Acc. switch rate:

0.001

0.001

0.2

Number of accidental candy type switches per day for a kid

☒ Relocation rate:

0.002

0.001

0.01

Number of relocations (moving from one town to another) per day for a kid


anylogic®
This model is
© AnyLogic North America
www.anylogic.com

▶

■

⌂

⌂

x1

⌂

⌂

⌂


Idle

⌂

⌂

⌂


The University of
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UNITED KINGDOM • CHINA • MALAYSIA

GenAI meets ABM

16



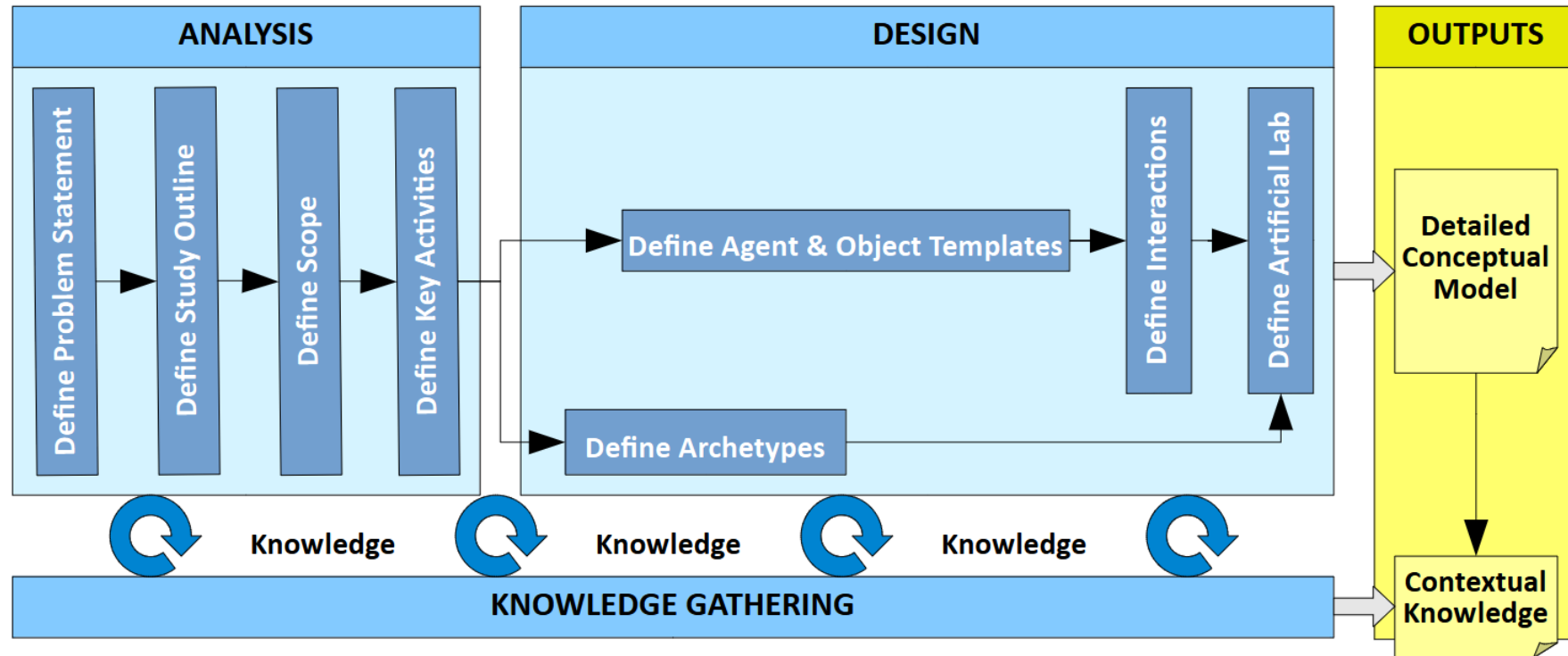


Using LLMs for ABM Design

[Exploring the Potential of Conversational AI Support for Agent-Based Social Simulation Model Design](#)

Conceptualising Agent-Based Models

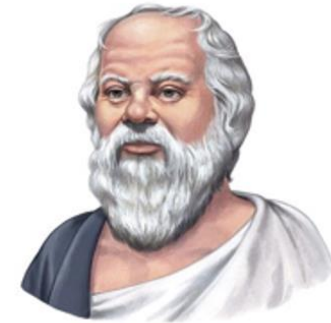
- The Engineering Agent-Based Social Simulation Framework



Adapted from Siebers & Klügl (2017)

The EABSS Framework

- Using mini focus groups
 - Group sizes of **4-5 participants** (including moderator) work best
 - Estimated time to get through the whole process: **4 hours** (or more)
- Confucius vs Socrates
 - Collaborative (inclusive) brainstorming
 - Mainly discussions (information gathering)
 - Debates only when needed
- Iterative process
 - Reuse of information from previous steps
 - Important to go forward and backwards



Engineering Agent-Based Social Simulations

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v4 (08/07/2024)

Brief Summary

What is it all about? What are successful model development strategies for complex social systems modelling? How does modelling work in large, collaborative, and multi-disciplinary projects in academia, non-governmental and governmental organisations, and industry? How do we derive new and general insights from modelling complex social systems? One solution is the use of agent-based modelling. Here, a complex system is represented by a collection of actors (individuals, households, firms, organisations, nations, etc.) that are programmed to follow some behaviour rules. System properties emerge from its constituent agent interactions when simulating scenarios, as simulation allows capturing the dynamics of a system over time. Anecdotal evidence suggests that the community of agent-based modellers partially suffers from a lack of structured and standardised ways for model development. To close this gap, we have created a model development framework, namely the Engineering Agent Based Social Simulation (EABSS) framework. It supports model development and model documentation in a structured way. Figure 1 shows a high-level overview of the latest version of the framework, EABSS-2. The latest news about the framework and extensive training resources can be found at the EABSS website (<https://www.cs.nott.ac.uk/~pszp/s/eabss-toolkit/>).

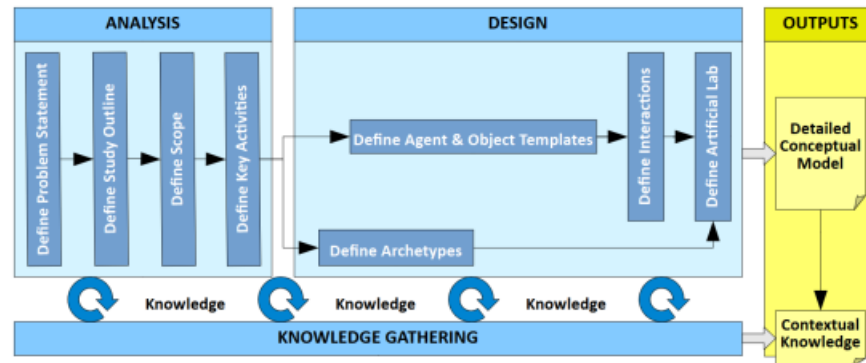


Figure 1: High level overview of the EABSS-2

How does it work? The EABSS is grounded in the concept of co-creation and ideas from Software Engineering. In addition it draws on elements of Kankainen's focus group approach to service design. The framework implicitly provides ground rules, which is something commonly done when working with children but often forgotten when working with grownups. These grounded rules are in line with De Bono's philosophy of parallel thinking, and state that people are going to listen to each other, and that people respect each other's point of view. To capture information it

Appendix 1: EABSS-2 Guidance (for moderators)

Small print **orange remarks** are meant to guide the focus group moderator regarding the re-use of information; **purple remarks** list the tools to be used in that particular step.

ANALYSIS

Define Problem Statement (also clarify terminology and come up with a common pool of term definitions)

- Clarify the "Purpose" of the model, if not provided by the client:
 - Study title
 - Broad theme(s) (Blue Sky; Toy; Organisation Studies; Social Studies; Operations Research; Operations Management; Economics; Ecology; ...)
 - Sub-theme(s) (Service Systems; Emergency Modelling; Organisational Cognition; Human/Natural Systems; Occupant behaviour; ...)
 - Study approach (level of abstraction [strategic ...]; purpose [exploratory ...]; driver [theory ...])
 - Study context (background information that provides a framework for understanding the research problem and its significance)
 - Study aim (a broad statement indicating the general purpose of your simulation study; consider title + themes + context)

Define Study Outline (also clarify terminology and come up with a common pool of term definitions)

- Define a list of "Objectives" (and constraints) to be fulfilled and/or "Hypotheses" to be tested (objectives: how you plan to achieve your aim; hypotheses: an attempt at explaining a phenomenon or the relationships between phenomena/variables in the real world)
- Define a list of "Experimental Factors" (parameters) to allow creating scenarios relevant to testing objectives and/or hypotheses (look at objectives/hypotheses to work these out)
- Define a list of "Responses" (outputs/statistics) for measuring if objectives have been achieved and to test if hypotheses should be accept/reject (look at objectives/hypotheses to work these out)

Define Scope (what do we need to represent to fulfil the aim; use "Context" words captured and "nouns" from the previous answers)

- List entities (key actors - represented by the role they play, and key objects) and concepts (key actors can also represent social/economic units, as for example families or firms)
- Indicate if these should be included/excluded in the model and justify your decision
- Use pre-defined table (COLUMNS: Category; Sub-Category; Element; Decision; Justification; CATEGORIES: Actor; Physical Environment; Social and Psychological Aspects; Misc)

Define Key Activities (actors come from scope table; key activities come from objectives/hypotheses and by creating user stories)

- Formulate user stories: As <actor>, I want to <what?> (so that <why?>)
- Assign key actors to relevant key activities (use cases)
- Use UML use case diagram

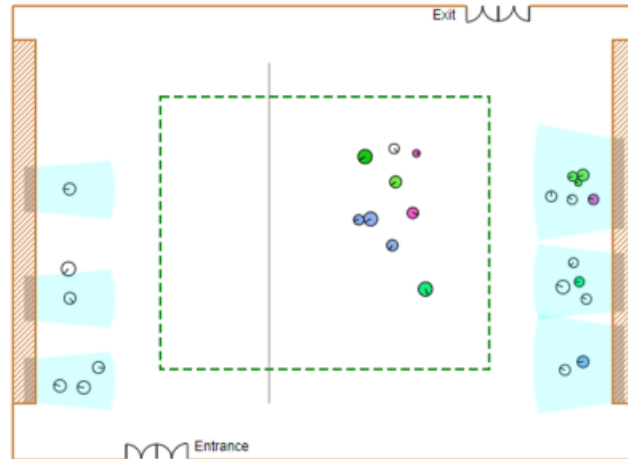


Appendix 2: Worked Example @ Adaptive Architecture

The information provided below was collected during several focus group sessions. The goal of the study was to come up with radical innovative ideas for incorporating AI artifacts in a museum exhibition hall. For capturing the information in a structured way, the EABSS framework was used. While the original version of the conceptional model was developed using pen and paper, the final version has been transcribed into a digital format to enhance readability.

In order to better grasp what we aim to develop in this worked example, we provide a preview of the model implementation. This did not exist before running the focus group sessions. It is based on the conceptual model resulting from the focus group sessions. To initiate the co-creation process, only the context was provided.

The implementation shows the museum exhibition hall layout and the interaction of visitors with the artefacts. There are two large screens on the left and right wall of the room, and several moving windows on the screens. Window size and position change depending on the number of visitors reading them and their movement. In the middle, we have a waiting area where visitors can relax. In the bottom left is an entrance where visitors are coming in at a specific rate, and in the top right is an exit where visitors leave the museum once they complete their journey through the interactive displays. Wall movement and visitor movement are based on the same principle, using the Social Force Model as a basis. Therefore, the wall moves with care, avoiding any obstacles, and optimising the space for visitors depending on the numbers on each side. Visitors are represented as coloured circles, where the same colours indicate groups of visitors, and the line represents the direction of view. The size of the circles denotes the body radius of the visitor.



ANALYSIS

Define Problem Statement

Study title

- Exploring Adaptive Architecture Design

Broad theme(s)

- Social Studies

Sub-theme(s)

- Occupant Behaviour

Study approach

- Strategic + Exploratory + Theory

Study context

- Adaptive architecture is concerned with buildings that are designed to adapt to their environments and their occupants. The purpose of this study is to explore Adaptive Architecture design in the context of a novel museum visit experience. In particular, we want to test ideas of adding two artificial intelligent artefacts to the exhibition hall to improve visitors' experience: (1) a large screen with a set of artificial intelligent proactive content windows that update position and size in response to movement and grouping of people in front of them, and (2) an artificial intelligent proactive partition wall that continuously monitors visitor movement and dynamically adapts its position in real-time to optimise the space available to the visitors on either side of the wall.

Study aim

- Explore the effect of incorporating artificial intelligent artefacts (content windows and partition wall) in a museum exhibition hall.

Focus Group Participants

- Modeller: Responsible for model design and implementation
- Adaptive Architecture Expert: Researcher in the field of Adaptive Architecture
- Simulation Expert: Expert in Agent-Based Social Simulation and the use of the EABSS framework
- Visitor: Represented indirectly by all participants, sharing their experience with museum visits

Define Study Outline

Objectives

1. Study the interaction of artificial intelligent content windows and visitors' movement
2. Study the interaction of an artificial intelligent partition wall and visitors' movement
3. Use the model to demonstrate to architects the idea of artificial intelligent artefacts

Hypotheses

1. A larger content window size has a positive effect on visitor engagement
2. Space availability has a positive effect on visitor engagement
3. Screens with artificial intelligent content windows attract viewers for longer

Experimental Factors

1. Visitor arrival rate
2. Initial number of content windows
3. Content window size
4. Length of partitioning wall (in relation to the length of the hall)
5. A subset of parameters of the underlying theoretical movement model

Conversational AI System (CAIs)

- AI that engages in natural dialogue, understanding, processing, and generating human-like language responses
- Key components:
 - NLP: Natural Language Processing (understanding what you mean)
 - NLG: Natural Language Generation (crafting coherent replies)
 - Transformer: Neural architecture powering modern CAIs
 - LLM: Large Language Models pre-trained on vast text corpora

The Research Question

- Can Conversational AI Systems like ChatGPT support the design of conceptual ABSS models, reducing need for expensive and logistically complex focus group sessions?
- Two core opportunities:
 - Idea Generator: Conduct a dry run through the EABSS framework using a CAIS to generate domain knowledge and modelling ideas with minimal upfront knowledge.
 - Virtual Stakeholder: Use role-play games with AI-powered 'virtual stakeholders' possessing domain knowledge, simulating co-creation discussions.

Prompt Engineering: The Art of Talking to AI

- Prompt engineering is crafting inputs that elicit accurate, relevant, and well-structured responses from a CAIS.
- Standard Prompt Template I used in this Project (Busch et al., 2023):
 - Context: Background information the model needs
 - Instructions: Clear, explicit task description
 - Examples: Show the desired output format
 - Constraints: Word limits, style, format rules
 - Ambiguity: Acknowledge and address uncertainty
 - Feedback Loop: Request self-review and refinement

Controlling AI Behaviour: Roles, Tones & Stochasticity

- Roles & Tones:
 - Defining a role shapes domain knowledge and vocabulary.
 - Example role: "Take on the role of a Sociologist with experience in ABSS"
 - Example tone: "Use a debating tone / scientific tone / inspirational tone"
- Temperature & Top_p:
 - Temperature: controls randomness. Low → deterministic | High → diverse/creative
 - Top_p (nucleus sampling): Low → focused | High → broader possibilities
- Same prompt but different Temp settings:
 - LOW: "Einstein's theory of relativity posits that the laws of physics are invariant across inertial frames..."
 - HIGH: "Einstein's relativity is like a trampoline made of starlight, where planets bounce, time melts like jelly..."

Prompt Design Patterns

- Inspired by Software Engineering design patterns: reusable solutions to recurring prompt design problems
 - General Pattern
 - Modules: Assign a role. Define terminology. Identify elements. Apply requirements. Specify further. Memorise in keys. Allow intervention.
 - Co-creation Pattern
 - Modules: Stakeholders debate. Debating tone. Random moderator question. Agree on outcomes. Rank criteria. Return to scientific tone. Memorise in keys. Allow intervention.
 - Table Pattern
 - Modules: Define table format. Define elements. Add requirements. Specify further. Organise by rules. Memorise in keys. Allow intervention.
 - Diagram Pattern
 - Modules: Generate Mermaid.js script. Use memorised info. Satisfy requirements. Increase complexity. Self-reflect & improve. Memorise in keys. Allow intervention.

Mimicking Fine-Tuning & Embedding Co-Creation

- Mimicking On-the-fly Fine-tuning
 - The EABSS framework's information reuse principle means each step feeds the next, incrementally training ChatGPT on your specific case study without API access.
- Simulating Co-creation with Virtual Stakeholders
 - Benefits
 - Reduces bias from a single 'general' viewpoint
 - AI generates novel random moderator questions each run
 - Useful when real stakeholders unavailable
 - Limitations
 - Cannot replicate authentic human critical thinking
 - Discussion is structured and artificial
 - No lived experience or genuine intuition

Problem Statement

- Display MD "Problem Statement". Render as 'Heading Level 3'. Only show rendered result| Take on the "role" of a "Sociologist" with experience in "Agent-Based Social Simulation". Memorise this role as {key-role1}. List memorised key-role1| Define the "topic" of the memorised key-studyType as "The goal of this study is to generate IDEAS for using ADAPTIVE ARCHITECTURE in futuristic MUSEUMS within an exhibition room that is visited by ADULTS and CHILDREN. The adaptive architecture consists of 2 kinds of artefacts: (1) 2 large wall-mounted SCREENS on which SMART CONTENT WINDOWS move with the visitors and (2) a SMART PARTITION WALL that creates a dynamic and flexible exhibition environment by continuously analysing visitor movement, making real-time decisions, and physically reconfiguring the space to optimise the experience for everyone. The adaptive architecture artefacts represent AI-DRIVEN INTELLIGENT OBJECTS.". Memorise this topic as {key-topic}. List memorised key-topic| Memorise "Exploratory" as {key-researchDesign}. List memorised key-researchDesign| Memorise "Social Studies" as {key-domain}. List memorised key-domain| Memorise "Human Behaviour" as {key-specialisation}. List memorised key-specialisation| Use an "inspirational tone". Define a novel and creative "context" for the memorised key-topic in 200 WORDS (if possible)| Use a "scientific tone". Memorise this context as {key-context}. List memorised key-context| Define 5 "stakeholders" for the memorised key-topic, to participate in a co-creation role-play game. Memorise these 5 stakeholders together with their personas (without names) as {key-stakeholders}. List memorised key-stakeholders.
- Explain how your answers differ for different memorised key-stakeholders. Be explicit for each identified stakeholder.
- Play a co-creation role-play game in which all the memorised key-stakeholders discuss with each other potential aims for the study considering the pros and cons. Use a "debating tone". The moderator focuses on 1 novel RANDOM question. Provide the question and the details of the controversial discussion| Agree on 2 potential aims that satisfy the view of all participating memorised key-stakeholders| Memorise these potential aims as {key-potentialAims}| Propose 3 criteria for ranking the 2 potential aims to support the decision which aim to carry forward| Use a "scientific tone".
- Define 5 "keywords" for the memorised key-studyType in the context of the memorised key-topic in the form of a comma-separated list. Memorise these 5 keywords as {key-keywords}. List memorised key-keywords| Use an "inspirational tone". Define the "title" for the memorised key-studyType in the context of the memorised key-topic in 12 WORDS (if possible). Memorise this title as {key-title}. List memorised key-title| Define the "aim" for the memorised key-studyType in the context of the memorised key-topic in 40 WORDS (if possible). CONSIDER the memorised key-potentialAims in your definition| Use a "scientific tone". Memorise this aim as {key-aim}. List the memorised key-aim.

Study Outline

- Display MD "Study Outline"| Render as 'Heading Level 3'. Only show rendered result| Define the term "objective" in the context of the memorised key-studyType in 1 concise sentence| Define the term "hypothesis" in the context of the memorised key-studyType in 1 concise sentence| Define the term "experimental factor" in the context of the memorised key-studyType in 1 concise sentence| Define the term "output" in the context of the memorised key-studyType in 1 concise sentence. List these 4 definitions.
- Play a co-creation role-play game in which all the memorised key-stakeholders discuss with each other potential ABSS objectives for the study considering the pros and cons. Use a "debating tone". The moderator focuses on 1 novel RANDOM question. Provide the question and the details of the controversial discussion| Agree on 4 potential ABSS objectives that satisfy the view of all participating memorised key-stakeholders| Memorise these potential ABSS objectives as {key-potentialObjectives}|

Model Scope

- Display MD "Model Scope"| Render as 'Heading Level 3'. Only show rendered result| Take on the "role" of a "Senior Software Developer" with experience in the "Unified Modelling Language". Memorise this role as {key-role2}. List memorised key-role2| Define the term "model scope" in the context of the memorised key-study in 1 concise sentence. Define the term "UML actor" in the context of the memorised key-uml in 1 concise sentence. List these 2 definitions.
- Play a co-creation role-play game in which all the memorised key-stakeholders discuss with each other potential ABSS UML actors for the study considering the pros and cons. Use a "debating tone". The moderator focuses on 1 novel RANDOM question. Provide the question and the details of the controversial discussion| Agree on 8 potential ABSS UML actors that satisfy the view of all participating memorised key-stakeholders| Memorise these potential ABSS UML actors as {key-potentialUMLActors}| Propose 3 criteria for ranking the 8 potential ABSS UML actors to support the decision which ABSS UML actors to carry forward| Use a "scientific tone".
- Define 4 "ABSS UML actors" as USERS OF THE SYSTEM described in the memorised key-topic. CONSIDER the memorised key-potentialUMLActors in your definitions. Memorise these 4 UML actors together with a persona description as {key-umlActors}. List memorised key-umlActors.
- Use TABLE format WITH "plaintext" and WITHOUT any code formatting. DO NOT use "
". IGNORE ALL space limitations| Define 15 "real-world elements" with relevance to the memorised key-topic. You ALWAYS must satisfy the following 7 requirements for defining real-world elements: 1) Consider what 'real-world elements' are needed to represent in the model scope and to satisfy the memorised key-aim. 2) ALL 4 memorised key-umlActors MUST BE REPRESENTED. 3) At least 2 Physical Environment elements MUST be present. At least 2 Social Aspect elements MUST be present. At least 2 Psychological Aspect elements MUST be present. At least 2 Misc elements MUST be present. 4) Consider the memorised key-context. 5) Consider all nouns in the conversation history. 6) Each element can only be in 1 category. 7) Social Aspect elements MUST describe theories of social behaviour. 8) Psychological Aspect elements MUST describe theories of psychological behaviour. Feel free to be creative and add your ideas. Categorise the 'real world elements' into Actors, Physical Environment, Social Aspects, Psychological Aspects, and Misc. TABLE MUST include 15 rows. Organise all 15 elements into categories and provide a brief explanation. Memorise these 15 elements and explanations as {key-explanations}| List the memorised key-topic relevant real-world elements in the form of table rows. Provide a column for Category. Provide a column for Sub-Category. Provide a column with the memorised key-explanations. Provide a column with concise justifications in ABOUT 25 WORDS. Memorise this table as {key-modelScope}.
- Use TABLE format WITH "plaintext" and WITHOUT any code formatting. DO NOT use "
". IGNORE ALL space limitations| Define 4 models for implementing elements of the memorised key-modelScope. Provide 1 social model AND 1 behavioural model AND 1 psychological model AND 1 technical model. Find relevant theoretical models in the SCIENTIFIC LITERATURE. Provide a full EXISTING UP-TO-DATE scientific paper (conference or journal) or book REFERENCE in HARVARD STYLE for each in a separate column. Memorise these 4 model details together with a description and the relevant reference as {key-implementationModels}.

Key Activities

- Display MD "Key Activities"| Render as 'Heading Level 3'. Only show rendered result| Define the term "user story" in the context of the memorised key-uml in 1 concise sentence| Define the term "use case" in the context of the memorised key-uml in 1 concise sentence. List these 2 definitions.
- Use TABLE format WITH "plaintext" and WITHOUT any code formatting. DO NOT use "
". IGNORE ALL space limitations| Define 2 "UML user stories" for each of the 4 memorised key-umlActors (Example: As an 'actor' I want 'action' so that 'achievement'). Memorise ALL 8 UML user stories as {key-umlUserStories}. Translate the memorised key-umlUserStories into UML use cases. Memorise ALL 8 UML use cases as {key-umlUseCases}. List ALL 8 memorised key-umlUserStories and ALL 8

The aim of this study is to explore the integration of adaptive architecture into futuristic museums to enhance visitor experience and educational value. By investigating the potential of smart content windows and AI-driven objects, the study seeks to create dynamic and engaging environments that cater to diverse audiences.

- **Museum Curator:** Responsible for overall museum design
 - The curator is concerned with the overall museum enthusiastic about innovation but cautious about risk. Therefore, they might focus on how the adaptive architecture can be integrated into the museum's design without overshadowing its exhibits. They would balance traditional and futuristic elements to ensure the museum remains accessible and engaging for all visitors.
- **Architect:** Designs the physical layout and structure of the museum
 - The architect is responsible for designing the physical space. They are interested in integrating technology seamlessly into the architecture to enhance the museum's functionality of the space. Their focus might be on how adaptive architecture can be incorporated into the museum's design without compromising its structural integrity. They might emphasize the need for flexible spaces that can adapt to different types of exhibits brought about by the adaptive architecture.
- **Educator:** Develops educational programs and activities for visitors
 - Educators are tasked with developing educational programs that leverage the museum's resources. They aim to enhance learning outcomes through interactive experiences. They would involve how adaptive architecture can be leveraged to create more engaging and personalized educational content tailored to different types of visitors.
- **Technologist:** Provides technical support for implementing digital experiences
 - Technologists provide technical support for implementing digital experiences. They are interested in innovation and eager to explore cutting-edge technologies. They would emphasize the need for robust infrastructure, including high-speed internet, secure data storage, and scalable software, and scalability to accommodate future technological advancements.
- **Visitor:** Represents the end-user experience
 - Visitors represent the end-user experience. They are interested in the museum's offerings during museum visits. Their perspective might influence the museum's design and programming. They would prioritize features that enhance their overall experience. They would prioritize features that enhance their overall experience, such as personalized content, interactive exhibits, and accessible environments that respond to their interests and needs. They would also prioritize accessibility and ease of navigation within the museum.

Category	Sub-Category	Explanation	Justification
Actor	Visitor	Represents individuals visiting the museum, including adults and children	Visitors are essential for simulating interactions with the museum environment and adaptive architecture
	Educator	Represents educators responsible for developing educational programs and activities for museum visitors	Educators play a key role in designing educational experiences and guiding visitor interactions
	Administrator	Represents the administrator responsible for the overall design and visitor experience of the museum	Administrators provide insights into the museum's vision and objectives, guiding the design of adaptive architecture
	Technician	Represents technician providing technical support for implementing adaptive architecture in the museum	Technicians ensure the feasibility and effectiveness of integrating technology into museum exhibits and spaces
Physical Environment	Exhibition Spaces	Represents the physical areas within the museum where exhibits are displayed	Understanding the layout and dynamics of exhibition
	Interactive Installations	Represents interactive installations and exhibits with the museum	
Social Aspects	Visitor Interactions	Represents social interactions among visitors, such group visits, conversations, and collaborative activities	
	Learning Communities	Represents communities of learners engaging in educational activities and programs within the museum	
Psychological Aspects	Engagement Levels	Represents the varying levels of engagement exhibited by visitors, ranging from passive observation to active exploration	
	Emotional Responses	Represents the emotional responses of visitors to exhibits and interactive elements within the museum	
Misc	Time	Represents the time spent by visitors in different areas of the museum	
	Spatial Layout	Represents the physical arrangement and layout of exhibits, interactive elements, and pathways within the museum	
	Safety	Represents emergency protocols and procedures for ensuring visitor and staff safety in case of emergencies	
	Feedback Mechanism	Represents the use of visitor feedback collection systems	
	Staff training	Represents training programs for educators to improve their skills and knowledge	

Use Case Diagram

```

graph LR
    Visitor((Visitor))
    Educator((Educator))
    Administrator((Administrator))
    ManageExhibits((Manage Exhibits))
    PlanWorkshop((Plan Workshop))
    CreateExhibit((Create Exhibit))
    SchedulesPresentation((Schedules presentation))
    UpdatesExhibit((Updates exhibit))
    ConductsWorkshop((Conducts workshop))

    Visitor -- "Manages exhibits" --> ManageExhibits
    Educator -- "Plans workshop" --> PlanWorkshop
    Administrator -- "Creates exhibit" --> CreateExhibit
    Visitor -- "Plans workshop" --> PlanWorkshop
    Visitor -- "Conducts workshop" --> ConductsWorkshop
    Administrator -- "Manages exhibits" --> ManageExhibits
    Administrator -- "Schedules presentation" --> SchedulesPresentation
    Administrator -- "Updates exhibit" --> UpdatesExhibit
    ManageExhibits -.-> PlanWorkshop
    ManageExhibits -.-> CreateExhibit
    ManageExhibits -.-> SchedulesPresentation
    PlanWorkshop -.-> ConductsWorkshop
    CreateExhibit -.-> UpdatesExhibit
    SchedulesPresentation -.-> UpdatesExhibit
  
```

```

graph TD
    Educator((Educator))
    Administrator((Administrator))
    Technician((Technician))
    Visitor((Visitor))
    Manager(( ))

    Manager -- "Manages exhibits" --> ManageExhibits[Manage Exhibits]
    Manager -- "Plans workshop" --> PlanWorkshop[Plan Workshop]
    Manager -- "Conducts workshop" --> ConductWorkshop[Conduct Workshop]
    Manager -- "Updates exhibit" --> UpdateExhibit[Update Exhibit]
    Manager -- "Creates exhibit" --> CreateExhibit[Create Exhibit]
    Manager -- "Schedules presentation" --> SchedulePresentation[Schedule Presentation]
    Manager -- "Performs maintenance" --> PerformMaintenance[Perform Maintenance]
    Manager -- "Monitors functionality" --> MonitorSystems[Monitor Systems]
    Manager -- "Monitors systems" --> MonitorSystems
    Manager -- "Resolves technical issues" --> ResolveTechnicalIssues[Resolve Technical Issues]

    Educator -- "Manages exhibits" --> ManageExhibits
    Educator -- "Plans workshop" --> PlanWorkshop
    Educator -- "Conducts workshop" --> ConductWorkshop

    Administrator -- "Updates exhibit" --> UpdateExhibit
    Administrator -- "Creates exhibit" --> CreateExhibit
    Administrator -- "Schedules presentation" --> SchedulePresentation

    Technician -- "Performs maintenance" --> PerformMaintenance
    Technician -- "Monitors functionality" --> MonitorSystems
    Technician -- "Monitors systems" --> MonitorSystems
    Technician -- "Resolves technical issues" --> ResolveTechnicalIssues

    Visitor -- "Explores exhibits" --> ExploreExhibits[Explore Exhibits]
    Visitor -- "Purchases ticket" --> PurchaseTicket[Purchase Ticket]
    Visitor -- "Receives guided tour" --> ReceiveGuidedTour[Receive Guided Tour]
    Visitor -- "Attends presentation" --> AttendPresentation[Attend Presentation]

    Manager -- "Diagnoses problems" --> ResolveTechnicalIssues
    Manager -- "Executes plan" --> ConductWorkshop
    Manager -- "Allocates resources" --> SchedulePresentation
    Manager -- "Guides participants" --> ParticipateInWorkshops[Participate in Workshops]
    Manager -- "Provides feedback" --> ProvideFeedback[Provide Feedback]
    Manager -- "Follows tour guide" --> AttendPresentation

    ManageExhibits -.-> MonitorSystems
    PlanWorkshop -.-> ConductWorkshop
    UpdateExhibit -.-> SchedulePresentation
    CreateExhibit -.-> SchedulePresentation
    SchedulePresentation -.-> ParticipateInWorkshops
    ExploreExhibits -.-> ParticipateInWorkshops
    PurchaseTicket -.-> ReceiveGuidedTour
    ReceiveGuidedTour -.-> AttendPresentation
  
```

Results: What Did ChatGPT Produce?

- High Quality
 - Usability: Script follows clear templates; minimal learning curve for EABSS users
 - Generality: Only topic description needs changing; script adapts to any case study
 - Pertinency: Output remained tightly connected to museum adaptive architecture topic throughout
 - Readability: Responses were coherent, well-structured, and readable by all stakeholders
 - Conformity: All 8 EABSS framework steps were addressed in the generated report
- Moderate Quality
 - Believability: Stakeholder debate was structured and credible, but clearly artificial
 - Originality: Several novel ideas emerged; some expected elements appeared due to LLM training bias
- After review, responses can be used as requirements for vibe-coding implementations



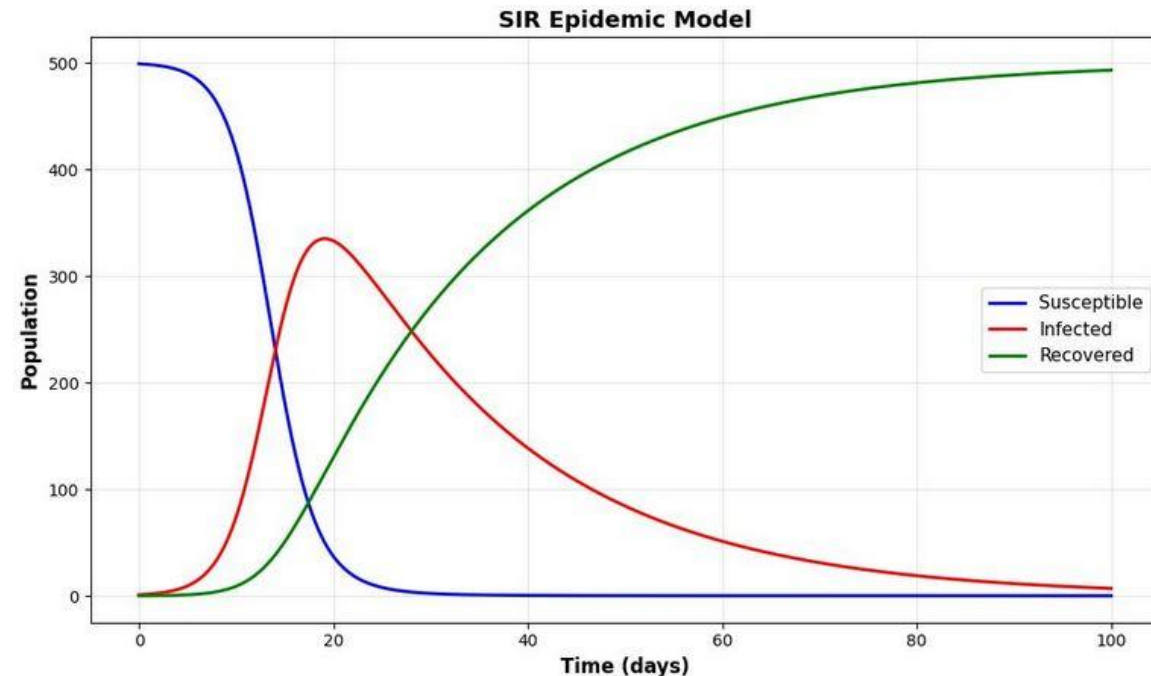
Generative Agent-Based Modelling

<https://people.cs.nott.ac.uk/pszps/abm.html#post11>

<https://people.cs.nott.ac.uk/pszps/abm.html#post12>

Use Case: The SIR Epidemic Model

- Imagine simulating a disease outbreak in a town with 500 residents. Each person decides their daily actions (go to work, stay home, or seek medical care) based on infection rates, health status, and personal circumstances.



Introducing Generative ABM (GABM)

Traditional ABM	Generative ABM
Pre-defined, deterministic rules	LLM-driven, context-aware decisions
Agents follow IF-THEN logic	Agents query an AI for guidance
Behaviour set at design time	Behaviour adapts to situation
Fast but lacks nuance	Captures human-like reasoning
Cannot reflect personal context	Unique persona per agent

Running LLMs Locally with KoboldCPP

- Why run locally
 - Privacy; Cost; Control
- Setup
 - Model: qwen2.5-1.5b-instruct-q6_k (1.5 GB) from Hugging Face
 - Hardware used: Intel i5 (2018) 8 GB RAM, 2 GB VRAM, Windows 10 (x64)

I wanted to learn how to do it!

Building the First Prototype (using Extreme Programming)

1. Simple Python Script: A few agents each with their own personality. LLM questions sent via API to KoboldCPP
2. Communication Verified: KoboldCPP running in one terminal, Python in another. First successful agent response received
3. SIR Model Integration: Agents in an SIR grid. Susceptible agents query the LLM each day: isolate or not?
4. Working Simulation: Agents make LLM-driven decisions. Visual state updates per simulated day. Logging system active

```
C:\Windows\System32\cmd.exe

Starting Agent-Based Simulation...
Make sure KoboldCpp is running on http://localhost:5001

=====
ROUND 1: You find a mysterious package at your door.
=====

You are Alice, an optimistic and helpful person. What are your thoughts?
I'm so glad I'm home, and I wonder what this mysterious package could be - it looks like it's been waiting for me!

You are Bob, a cautious and analytical person. What are your thoughts?
I'll examine the package for any identifying information and then call the authorities to report it.

You are Charlie, a creative and spontaneous person. What are your thoughts?
I'm so curious, I just have to open this mysterious package and see what's inside!

=====
ROUND 2: You're invited to explore an abandoned building.
=====

You are Alice, an optimistic and helpful person. What are your thoughts?
"Let's start our adventure by carefully checking the building's foundation and making sure it's safe to enter!"

You are Bob, a cautious and analytical person. What are your thoughts?
I'll accept the invitation, but I'll make sure to bring my flashlight and wear sturdy shoes to ensure my safety.

You are Charlie, a creative and spontaneous person. What are your thoughts?
"I can't wait to find out what secrets and surprises this old place has in store for us."
```

How LLM Decision-Making Works

- Gather context: Count nearby S, I, R agents within observation radius
- Build prompt
 - Personalised prompt: persona + neighbourhood stats + question
 - Parse response: POST request to KoboldCPP API — awaits YES/NO response
- Parse response: Keyword match in first 50 chars; fallback rule if unclear
- Apply behaviour: Isolate → move at 20% speed; not isolate → normal speed
- Each susceptible agent re-evaluates every single simulated day with updated neighbourhood data

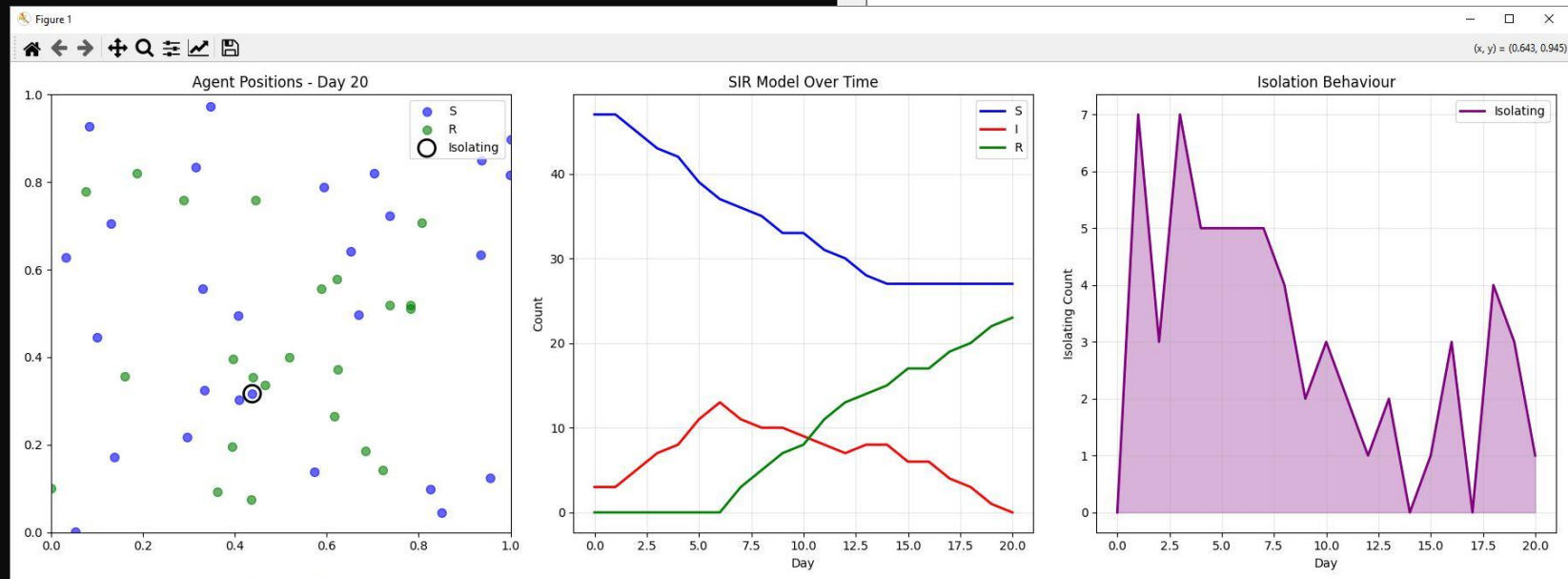
A Real LLM Interaction

- Prompt send to LLM
 - "Cautious healthcare worker, isolates early. Nearby: 7 people, 2 infected. Should you isolate? Reply: YES/NO then reason briefly."
- LLM response
 - "YES. As a healthcare worker, I understand transmission risks. Two infections nearby is significant, and early isolation prevents spread."
- Result
 - Agent isolates → moves at 20% normal speed. Skipped in disease spread calculations for that step. Decision is re-evaluated each day with fresh neighbourhood data.

```
C:\Windows\System32\cmd.exe - python llm-driven-abm-sir_v08-error-stats-fixed-advanced.py

=====
SIMULATION STATISTICS SUMMARY
=====

Simulation Overview:
  Days simulated: 20
  Total agents: 50
  Final S: 27, I: 0, R: 23
Runtime Statistics:
  Total simulation time: 37.93 minutes
  Average time per day: 1.90 minutes
  Total LLM processing time: 37.89 minutes
  Average LLM request time: 0.06 minutes
LLM Communication Statistics:
  Total requests: 688
  Successful: 688 (100.0%)
  Errors: 0 (0.0%)
  Default decisions: 0 (0.0%)
  Response Types:
    YES: 63 (9.2%)
    NO: 625 (90.8%)
    UNCLEAR: 0 (0.0%)
    ERROR: 0 (0.0%)
Decision Statistics:
  Total decisions made: 688
  Decision failures: 0
  Decision failure rate: 0.0%
  Isolation decisions: 63 YES, 625 NO
  Overall isolation rate: 9.2%
Persona Decision Patterns:
  Elderly with health concerns, very cautious
  Sceptical of warnings, values freedom
  Data-driven scientist
  Social person, finds isolation difficult
  Essential worker balancing safety and economics
  Cautious healthcare worker, isolates early
  Young optimist, values social connections
  Parent balancing work and family safety
Connection Statistics:
  Max concurrent requests: 1
  Max retries per request: 2
  Request timeout: 30s
=====
Simulation complete!
```



The "It Looks Right, But Is It?" Problem

- Contradictory example from the logs
 - Details submitted: "Social person, finds isolation difficult. Nearby: 4 people, 0 infected."
 - LLM response (contradictory!): "NO. The individual is a nearsighted person who struggles with social interactions..."
- General Issues
 - Ignored actual persona (social person $\neq$ nearsighted)
 - 0% infection rate mentioned, yet reasoning was still garbled
 - Many decisions triggered the fallback rule due to errors
 - Fluent language masked underlying incoherence
- LLMs present fabrications with confidence
 - Always validate at micro AND macro level

LLM-Augmented Agent-Based Modelling

 Cornell University

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[Submitted on 8 Jul 2025]

Large Language Models for Agent-Based Modelling: Current and possible uses across the modelling cycle

Loïs Vanhée, Melania Borit, Peer-Olaf Siebers, Roger Cremades, Christopher Frantz, Önder Gürçan, František Kalvas, Denisa Reshef Kera, Vivek Nallur, Kavin Narasimhan, Martin Neumann

The emergence of Large Language Models (LLMs) with increasingly sophisticated natural language understanding and generative capabilities has sparked interest in the Agent-based Modelling (ABM) community. With their ability to summarize, generate, analyze, categorize, transcribe and translate text, answer questions, propose explanations, sustain dialogue, extract information from unstructured text, and perform logical reasoning and problem-solving tasks, LLMs have a good potential to contribute to the modelling process. After reviewing the current use of LLMs in ABM, this study reflects on the opportunities and challenges of the potential use of LLMs in ABM. It does so by following the modelling cycle, from problem formulation to documentation and communication of model results, and holding a critical stance.

Comments: 18 pages, including 2 pages of appendix, accepted for publication at the Social Simulation Conference 2025 ([this https URL](#))

Subjects: **Multiagent Systems (cs.MA)**

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Exploring the dynamics of social systems

Image by Gemini Flash 2.5 (06/2025)

This is my blog on Agent-Based Modelling (ABM). It is simply a webpage I update from time to time. There is no subscription option. This blog is intended as a space for sharing news, discussions, and information related to my research on Computational ABM.

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Please note: These posts do not have any scientific rigour. They are simply exploratory pieces.

- [22/12/2025: 16 Special Issue on 'Generative AI meets Agent-Based Modelling and Simulation'](#)
- [22/12/2025: 15 LLM4ABM Special Interest Group \(SIG\) Established as an ESSA SIG](#)
- [14/12/2025: 14 LLM Support for Creating RAT-RS Reports \(2/2\): Quality & Reliability](#)
- [06/12/2025: 13 LLM Support for Creating RAT-RS Reports \(1/2\): Feasibility](#)
- [01/11/2025: 12 The Generative ABM Experiment \(1/3\): From Concept to First Prototype](#)
- [12/10/2025: 11 From Rules to Reasoning: Engineering LLM-Powered Agent-Based Models](#)

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